

Nathan A. Saffold

Fermi National Accelerator Laboratory
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EDUCATION

Columbia University, New York, NY

Ph.D., Physics 2021

M.Phil., Physics 2018

Williams College, Williamstown, MA

B.A., Astrophysics 2014

AWARDS AND GRANTS

Lederman Fellowship 2021–present

Fermilab Detector R&D New Initiatives 2024–2025

Columbia University Provost Diversity Fellowship 2015–2021

UC Berkeley Summer Research Opportunity Program (SROP) Fellow 2013

RESEARCH EXPERIENCE

Fermi National Accelerator Laboratory October 2021–present

Lederman Postdoctoral Fellow

Leading the development of the DarkNESS mission, a CubeSat dark matter search and technology demonstrator for the space qualification of skipper-CCDs:

- Characterize and calibrate skipper-CCDs for the DarkNESS detector module. Commission test chamber and oversee calibration data-taking process. Tune skipper-CCD operating parameters to achieve optimal noise and charge transfer performance.
- Develop in-flight analysis software to convert raw images into high-level data products for downlink to ground.
- Forecast DarkNESS science reach based on laboratory prototype testing.
- Responsible for coordinating weekly meetings with scientific and engineering staff.

Developing novel imaging sensor technologies and laboratory testing techniques at Fermilab CCD lab

- Supervising postdoc and graduate student researchers on the development of SiSeRO-CCDs for near-infrared and X-ray detection.
- Helped develop and apply automated sensor calibration technique using Bayesian optimization to characterize and optimize skipper- and SiSeRO-CCD operation.
- Developed a low-power instrument concept for mapping the lunar water content via epithermal neutrons using boron-coated silicon imagers.

Data analysis contributor for the SENSEI experiment, a direct detection experiment searching for sub-GeV dark matter with 100 g of skipper-CCDs

- Contributed to commissioning of SENSEI@SNOLAB detector, including the mapping of pixel defects in the CCD array and analyzing high-energy events to estimate the background rate.
- Led effort to optimize masking parameters for SENSEI@SNOLAB analysis before unblinding the first science run.
- Performed analysis cross-checks on the unblinded dataset, including fitting dark matter models to the measured charge spectrum.

Detector commissioning for the Oscura experiment, a next-generation search for sub-GeV dark matter with a 10 kg array of skipper-CCDs:

- Performed first measurements of dark current in immersion-cooled skipper-CCDs.
- Studied the scintillation light produced in various immersion cooling liquids.

Columbia Astrophysics Laboratory, Columbia University

June 2016 - August 2021

Graduate Research Assistant with Prof. Chuck Hailey

Contributed to detector development and sensitivity studies for the GAPS experiment, a balloon-borne indirect dark matter search utilizing low-energy antideuterons:

- Performed detector design and fabrication studies for the GAPS lithium-drifted silicon (Si(Li)) tracking system. Led the R&D effort to develop and optimize a passivation protocol for Si(Li) detectors.
- Led simulation effort to determine the GAPS sensitivity to detect antihelium-3 nuclei using GEANT4.
- Conducted GEANT4 simulations of antiproton, antideuteron, and antihelium annihilations in different target materials to benchmark annihilation products in new GEANT4 releases.
- Participated in scientific balloon flight at NASA CSBF facility to validate GAPS thermal cooling system.
- Performed spectral and imaging analysis of the eastern lobe of SS433 with NuSTAR.

UC Berkeley Space Sciences Laboratory

Summer 2013

Summer Research Assistant on COSI with Prof. Steve Boggs

TEACHING AND MENTORING EXPERIENCE

Saturday Morning Physics Co-Director, Fermilab

2023 - present

In charge of organizing and running Saturday Morning Physics, an outreach program that provides a semi-annual series of eleven lectures to high school students aimed to enhance their understanding and appreciation of modern physics. Responsible for scheduling speakers, assigning tasks to support staff, and hosting the lectures, including introducing the speakers and facilitating questions from the students. Present semi-annual lecture on dark matter to students.

Research Mentor, Fermilab

2021 - present

Mentor and develop research programs for undergraduate, graduate, and postdoctoral researchers from UChicago and Fermilab's SIST and SULI internship programs on skipper-CCD R&D projects:

Kodjo Akouedjinoude (2022-2023), Terry Kim (2023-2024), Samriddhi Bhatia (2024), Graydon Schulze-Kalt (2023-present), Joseph Noonan (2024-present), Rodrigo Spinola E. Castro (2025-present), Julian Cuevas-Zepeda (2024-present).

Columbia Astrophysics Laboratory, Columbia University

2017 - 2021

Mentored undergraduate and postbaccalaureate researchers on various GAPS projects:

Evan Martinez (2017-2020), Amani Garvin (2017-2018), Brian Smallshaw (2017-2018), Cianci Melo-Carrillo (2019), Raag Agrawal (2017), Gabriel Brown (2017), Gabriel Bridges (2019) Derik Kraych (2019), Quan Gan (2019), Emma Brann (2019), Yve Schutt (2019), Isaac Pope (2019-2020), Kanav Kalucha (2019-2020), Chelsea Rodriguez (2019-2020).

Graduate Teaching Fellow, Columbia University

2015–2018

Taught laboratory courses for General Physics I and II. Lectured for introductory engineering laboratory course. Served as Preceptor for Columbia Physics department, managing and instructing early-stage graduate teaching fellows.

LIST OF PUBLICATIONS

Peer-Reviewed Journal Articles

- P. Alpine, N. Saffold, et al., *DarkNESS: a skipper-CCD nanosatellite for dark matter searches*, **Adv. Space Res.** 76, 8 (2025).
- The SENSEI Collaboration, *SENSEI at SNOLAB: Single-Electron Event Rate and Implications for Dark Matter*, **Phys. Rev. Lett.** 134, 16 (2025).
- The SENSEI Collaboration, *SENSEI: First Direct-Detection Results on sub-GeV Dark Matter from SENSEI at SNOLAB*, **Phys. Rev. Lett.** 134, 011804 (2025).
- DAMIC, DAMIC-M and SENSEI Collaborations, *Confirmation of the spectral excess in DAMIC at SNOLAB with skipper CCDs*, **Phys. Rev. D** 109, 062007 (2024).
- The SENSEI Collaboration, *Search for Millicharged Particles Produced in the NuMI Beam*, **Phys. Rev. Lett.** 133, 071801 (2024).
- M. Xiao et al., *Large-Scale Detector Testing for the GAPS Si(Li) Tracker*, **IEEE Trans. Nucl. Sci.** 70, 2125–2133 (2023).
- Oscura Collaboration, *Searching for millicharged particles with 1 kg of Skipper-CCDs using the NuMI beam at Fermilab*, **JHEP** 02, 072 (2024).
- Oscura Collaboration, *Skipper-CCD Sensors for the Oscura Experiment: Requirements and Preliminary Tests*, **JINST** 18, P08016 (2023).
- S. Safi-Harb et al., *Hard X-ray emission from the eastern jet of SS 433 powering the W50 “Manatee” nebula: Evidence for particle re-acceleration*, **ApJ** 935, 163 (2022).
- The GAPS Collaboration, *Sensitivity of the GAPS experiment to low-energy cosmic-ray antiprotons*, **Astropart. Phys.** 145, 102791 (2023).
- A. Botti et al., *Constraints on the electron-hole pair creation energy and Fano factor below 150 eV from Compton scattering in a Skipper-CCD*, **Phys. Rev. D** 106, 072005 (2022).
- M. Kozai et al., *Statistical investigation of the large-area Si(Li) detectors mass-produced for the GAPS experiment*, **Nucl. Instrum. Meth. A** 1034, 166820 (2022).
- R. Munini et al., *The antinucleus annihilation reconstruction algorithm of the GAPS experiment*, **Astropart. Phys.** 133, 102640 (2021).

- N. Saffold et al., *Cosmic antihelium-3 nuclei sensitivity of the GAPS experiment*, **Astropart. Phys.** 130, 102580 (2021).
- N. Saffold et al., *Passivation of Si(Li) detectors operated above cryogenic temperatures for space-based applications*, **Nucl. Instrum. Meth. A** 997, 165015 (2021).
- M. Kozai et al., *Developing a mass-production model of large-area Si(Li) detectors with high operating temperatures*, **Nucl. Instrum. Meth. A** 947, 162695 (2019).
- F. Rogers et al., *Large-area Si(Li) detectors for X-ray spectrometry and particle tracking in the GAPS experiment*, **JINST** 14, P10009 (2019).
- K. Perez et al., *Fabrication of low-cost, large-area prototype Si(Li) detectors for the GAPS experiment*, **Nucl. Instrum. Meth. A** 905, 12–21 (2018).

Conference Proceedings and Technical Reports

- N. Saffold et al., *The DarkNESS mission: probing dark matter with a Skipper-CCD satellite observatory*, **Proc. of XVIII Int. Conf. on Topics in Astroparticle and Underground Physics (TAUP2023)**, PoS(TAUP2023) 062 (2023).
- P. Adari et al., *EXCESS workshop: Descriptions of rising low-energy spectra*, **SciPost Phys. Proc.** 9, 001 (2022).

Preprints and White Papers

- A. Botti, et al., *Light-tight skipper-CCDs for X-ray detection in space*, submitted to **PASP**.
- J. Cuevas-Zepeda, N. Saffold et al., *High-efficiency, low-noise X-ray detection with thick fully-depleted p-channel SiSeRO-CCDs*, in prep.
- P. Alpine, N. Saffold et al., *X-ray characterization of fully-depleted p-channel Skipper-CCDs for the DarkNESS mission*, in prep for journal submission.
- I. Bloch et al., *SENSEI: A Search for Diurnal Modulation in sub-GeV Dark Matter Scattering*, arXiv:2510.20889, submitted to PRL.
- J. Cuevas-Zepeda et al., *Automated sensor calibration with machine learning*, arXiv:2509.04367, submitted to JINST.
- J. Cuevas-Zepeda et al., *A compact, low-power epithermal neutron counter for lunar water detection*, arXiv:2509.21661, submitted to **JINST**.
- The Oscura Collaboration, *The Oscura Experiment*, arXiv:2202.10518.
- P. von Doetinchem et al., *Cosmic-ray antinuclei as messengers of new physics: status and outlook for the new decade*, **JCAP** 08, 035 (2020).

SELECT PRESENTATIONS AND SEMINARS

- *Demonstrating skipper-CCDs in space with DarkNESS*
Talk presented at the Scientific Detector Workshop 2025, Canberra, Australia, October 2025.
Received “Most Original Idea” award.
- *DarkNESS: probing dark matter from Low Earth Orbit with skipper-CCDs*
Talk presented at UCLA Dark Matter 2025, Los Angeles, CA, March 2025.
- *Flash Talk: Cosmology and Dark Matter*
Talk presented at the Conference for Undergraduate Women and Gender Minorities in Physics 2025 (CU*IP), Batavia, Illinois, January 2025.
- *The DarkNESS mission: searching for dark matter with a skipper-CCD satellite observatory*
Talk presented at the 15th International Workshop on the Identification of Dark Matter (IDM2024), L’Aquila, Italy, July 2024.
- *Sub-GeV Dark Matter searches with SENSEI*
Talk presented at the International Workshop on Double Beta and Underground Science (DBD23), Waikoloa, Hawaii, December 2023.
- *The DarkNESS mission: probing dark matter with a skipper-CCD CubeSat.*
Talk presented at XVIII International Conference on Topics in Astroparticle and Underground Physics (TAUP2023), Vienna, Austria, August 2023.
- *Sub-GeV Dark Matter searches with SENSEI*
Talk presented at APS April Meeting, Minneapolis, Minnesota, December 2023.
- *The Oscura Experiment – Searching for low-mass dark matter with a very large array of skipper-CCDs*
Talk presented at 14th International Conference on the Identification of Dark Matter (IDM2022), Vienna, Austria, July 2022.
- *The Oscura Experiment – Searching for low-mass dark matter with a very large array of skipper-CCDs*
Talk presented at 7th Symposium on Neutrinos and Dark Matter in Nuclear Physics (NDM22), Asheville, North Carolina, May 2022.
- *The General Antiparticle Spectrometer – Searching for Cosmic-ray Antihelium*
Talk presented at Committee on Space Research 2021 (COSPAR2021), Sydney, Australia, January-February 2021.
- *Passivation of Si(Li) Detectors operated above Cryogenic Temperatures for Space Applications*
Poster presented at IEEE Nuclear Science Symposium 2020, Boston, Massachusetts, November 2020.